

Zanzirangi System Architecture

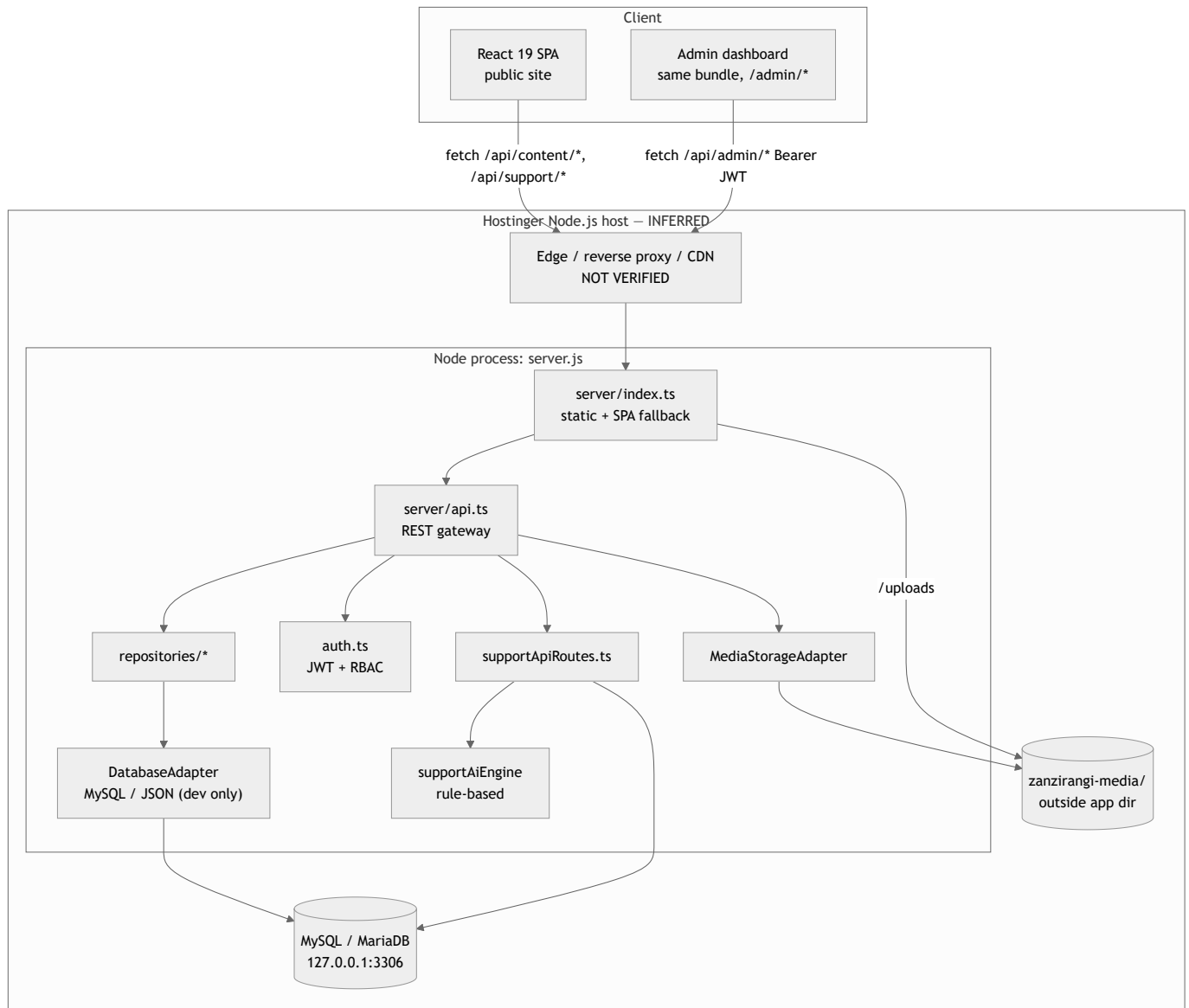
Audit date	2026-10-01
Revision	main @ 2379f82 + uncommitted working tree
Companion	ZANZIRANGI_PROJECT_BRIEF.md
Labels	VERIFIED · PARTIALLY VERIFIED · INFERRED · NOT VERIFIED · NOT FOUND · PLANNED

This document describes the architecture **as implemented**. Hosting-side facts (DNS, SSL, CDN, Node version on host) could not be verified from the repository and are marked accordingly.

1. Architectural Style

A **single-process, same-origin monolith** (VERIFIED):

- One Express 5 application (`server/index.ts` , bundled to `server.js`) serves:
 1. the compiled React SPA (`dist/`) and pre-rendered route HTML,
 2. uploaded media (`/uploads`),
 3. a JSON REST API (`/api/*` , defined in `server/api.ts` and `server/supportApiRoutes.ts`).
- The same `apiApp` is mounted into the Vite dev server in development (`vite.config.ts:11-22`), so dev and prod share one API implementation.
- No microservices, queues, caches (Redis), background workers or external APIs are used by the server.



2. Frontend

Aspect	Implementation	Label / Evidence
Framework	React 19 + TypeScript	VERIFIED <code>package.json</code>
Build tool	Vite 6, <code>@vitejs/plugin-react</code> , Tailwind CSS 4 plugin	VERIFIED <code>vite.config.ts</code>
UI system	Tailwind utility classes, custom CSS variables (<code>src/index.css</code>), separate admin theme (<code>src/admin/admin-theme.css</code> , untracked), <code>lucide-react</code> icons, motion animations, Google Fonts (Cormorant Garamond, Plus Jakarta Sans)	VERIFIED
Routing	Custom: <code>currentPath</code> state from <code>window.location.pathname</code> , <code>history.pushState</code> , <code>popstate</code> listener (<code>src/App.tsx:188-370</code>). Admin = any path beginning <code>/admin</code> ; tab = <code>/admin/<tab></code>	VERIFIED
State management	React <code>useState/useEffect/useMemo</code> in <code>App.tsx</code> , props drilling; <code>ThemeContext</code> ; <code>localStorage</code> for theme, language, admin token, chat visitor id	VERIFIED
Data access	<code>src/services/contentApi.ts</code> (public + admin content), <code>authApi.ts</code> , <code>supportApi.ts</code> ; static fallback data in <code>src/data/*</code>	VERIFIED

Aspect	Implementation	Label / Evidence
i18n	8 languages; static UI dictionaries + CMS translations from <code>/api/content/translations/:lang</code> applied by <code>src/i18n/cmsTranslations.ts</code> (untracked)	VERIFIED
Code splitting	None — one 1.73 MB JS chunk including admin	VERIFIED (build run 2026-10-01)
SEO runtime	Per-route <code><title></code> , meta, OG, canonical updated in <code>App.tsx:382-443</code> ; <code>scripts/generate_routes.js</code> copies <code>index.html</code> per route with route-specific meta	VERIFIED

Public routes

`/`, `/villas`, `/dining`, `/experiences`, `/safari`, `/about`, `/contact`, `/privacy`, `/terms`. Any other non-admin path renders the shell without content (no 404 view).

Admin views (`/admin/<tab>`)

`dashboard`, `support`, `homepage`, `pages`, `transfers`, `whystay`, `dining`, `experiences`, `safari`, `global`, `translations`, `admin-access`, `rooms`, `gallery`, `videos`, `facilities`, `testimonials`, `contact`, `seo`, `media`, `settings` (`src/App.tsx:518-563`). Unauthenticated → `AdminLogin`. Navigation hidden per permission (`AdminLayout.tsx:81-92`); authorisation enforced server-side.

3. Backend

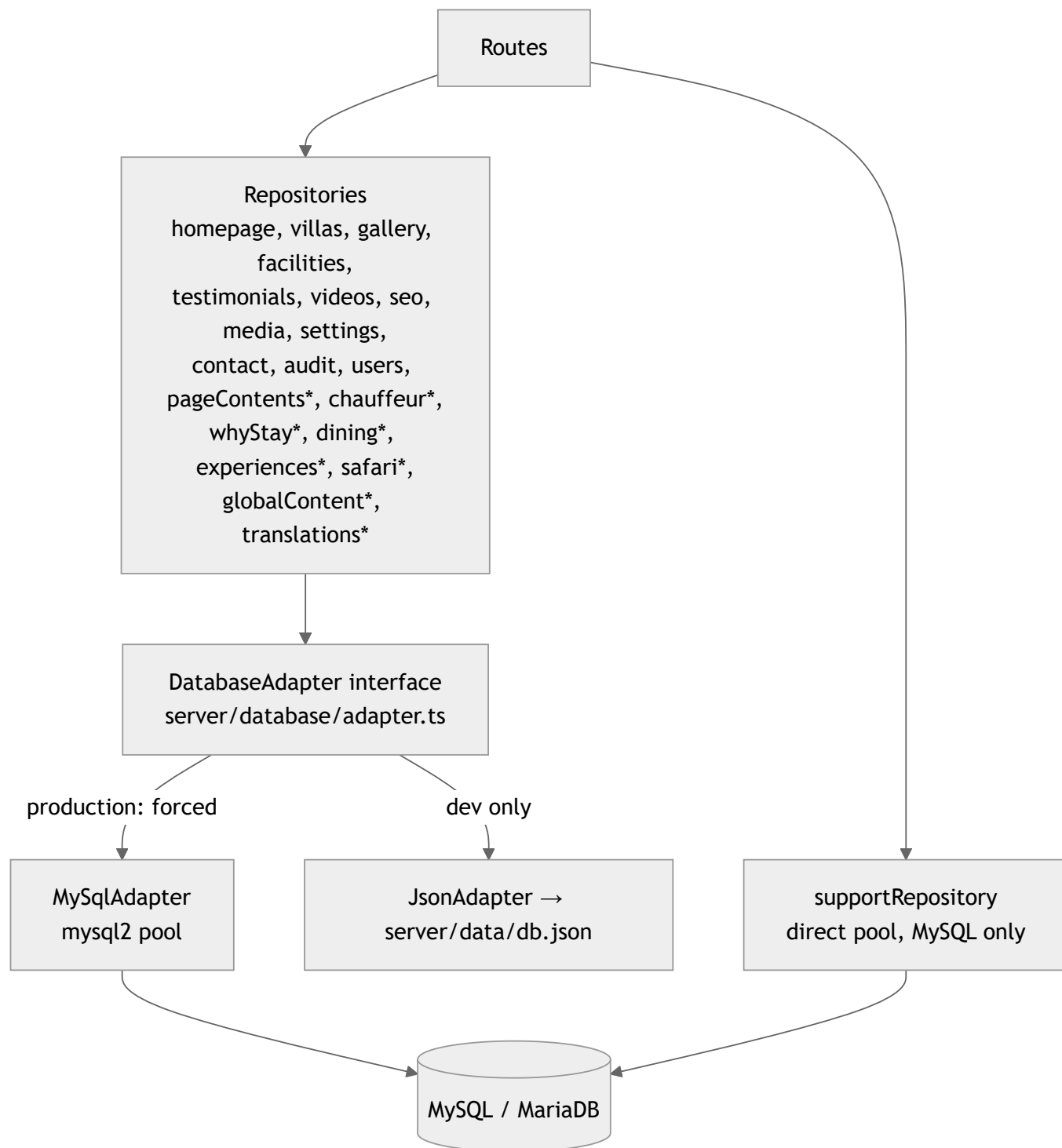
Aspect	Implementation	Evidence
Runtime	Node.js (<code>engines >=20</code>); local audit machine Node 24.16; CI Node 22; host version NOT VERIFIED	<code>package.json: 5-7</code>
Framework	Express 5	<code>package.json</code>
Entry	<code>server/index.ts</code> → esbuild → <code>server.js</code> (ESM, packages external)	<code>package.json: 15</code>
Bind	<code>`0.0.0.0:\${PORT}</code>	
Startup lifecycle	validate env (fail fast in prod) → listen → connect DB with retry/backoff (5s × attempt, max 60s) → <code>/api/health</code> reports 503 until ready	<code>server/index.ts:11-139</code>
Shutdown	SIGTERM/SIGINT → close HTTP → close pool → exit; 10 s forced-exit timer; <code>uncaughtException</code> triggers shutdown	<code>server/index.ts:141-178</code>
API architecture	REST/JSON, <code>{ success, data, error }</code> envelope; JSON 404 and error handler for <code>/api/*</code>	<code>server/api.ts:1641-1659</code>
Middleware order (API)	security headers → CORS (credentials) → cookie-parser → JSON body ($\approx 1.4\times$ upload limit) → urlencoded → routes → 404 → error handler	<code>server/api.ts:53-74</code>
Static serving	<code>/uploads</code> (persistent dir + legacy <code>./uploads</code>) with <code>nosniff</code> , <code>sandbox CSP</code> , <code>no-cache</code> ; <code>/assets</code> immutable; <code>dist/</code> with HTML <code>no-cache</code> ; SPA fallback serving pre-rendered <code>dist/<route>/index.html</code> when present	<code>server/index.ts:32-95</code>
Authentication	<code>server/auth.ts</code> (see §5)	VERIFIED
AI/Support engine	<code>server/services/supportAiEngine.ts</code> — regex rules → knowledge-base substring match → keyword FAQ → human hand-off; fixed confidence values; no external LLM	VERIFIED

API surface (VERIFIED — `server/api.ts`, `server/supportApiRoutes.ts`)

Group	Endpoints	Guard
Health	<code>GET /api/health</code>	public

Group	Endpoints	Guard
	GET /api/health/database	superadmin
Auth	POST /api/auth/login (rate-limited), POST /api/auth/logout, GET /api/auth/me	public / public / JWT
Public content (GET)	/api/content/ homepage, villas, gallery, facilities, testimonials, videos, seo, contact, settings, pages, pages/:id, home-sections, translations/:lang, chauffeur, whystay, dining, dining/categories, experiences, safari, global	public, read-only
Admin content	homepage, contact-info, villas[:id], gallery[:id], facilities[:id], testimonials[:id], videos, seo, media[:id], settings, pages[:id], translations/:lang, home-sections, chauffeur, whystay, dining, dining/categories[:id], experiences[:id], safari[:id], global	JWT + module permission
Media upload	POST /api/admin/media/upload	JWT only (no media permission check)
Users	GET/POST /api/admin/users, GET/PUT /api/admin/users/:id, POST .../disable, .../enable, .../reset-password	superadmin
Dashboard / audit	GET /api/admin/dashboard-stats (dashboard perm), GET /api/admin/audit-logs (superadmin)	
Support – visitor	POST /api/support/conversation, GET /conversation/:id, POST /conversation/:id/messages, GET /conversation/:id/poll	visitor_id ownership check; no rate limit
Support – admin	/api/support/admin/ conversations[:id], .../messages, .../status, .../suggested-reply, knowledge-base[:id], analytics	JWT + support

4. Data Layer

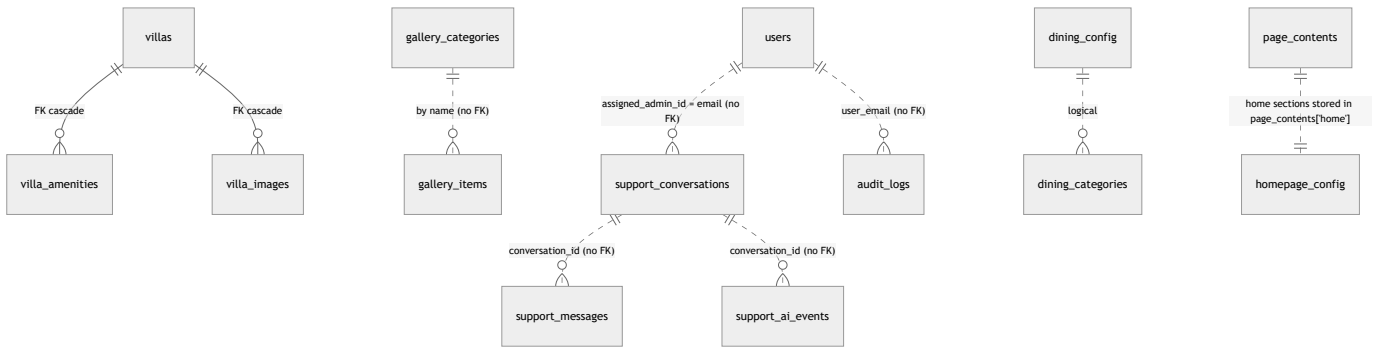


* = untracked (uncommitted) repository files.

Aspect	Detail	Label
Engine	MySQL-compatible, InnoDB, utf8mb4	VERIFIED (migrations); MySQL vs MariaDB on host NOT VERIFIED
Query layer	Raw mysql2/promise with ? placeholders; no ORM; no user-controlled interpolation found	VERIFIED
Pool	connectionLimit (default 10), keep-alive, 15 s connect timeout, no TLS option	VERIFIED server/database/connection.ts:43-55

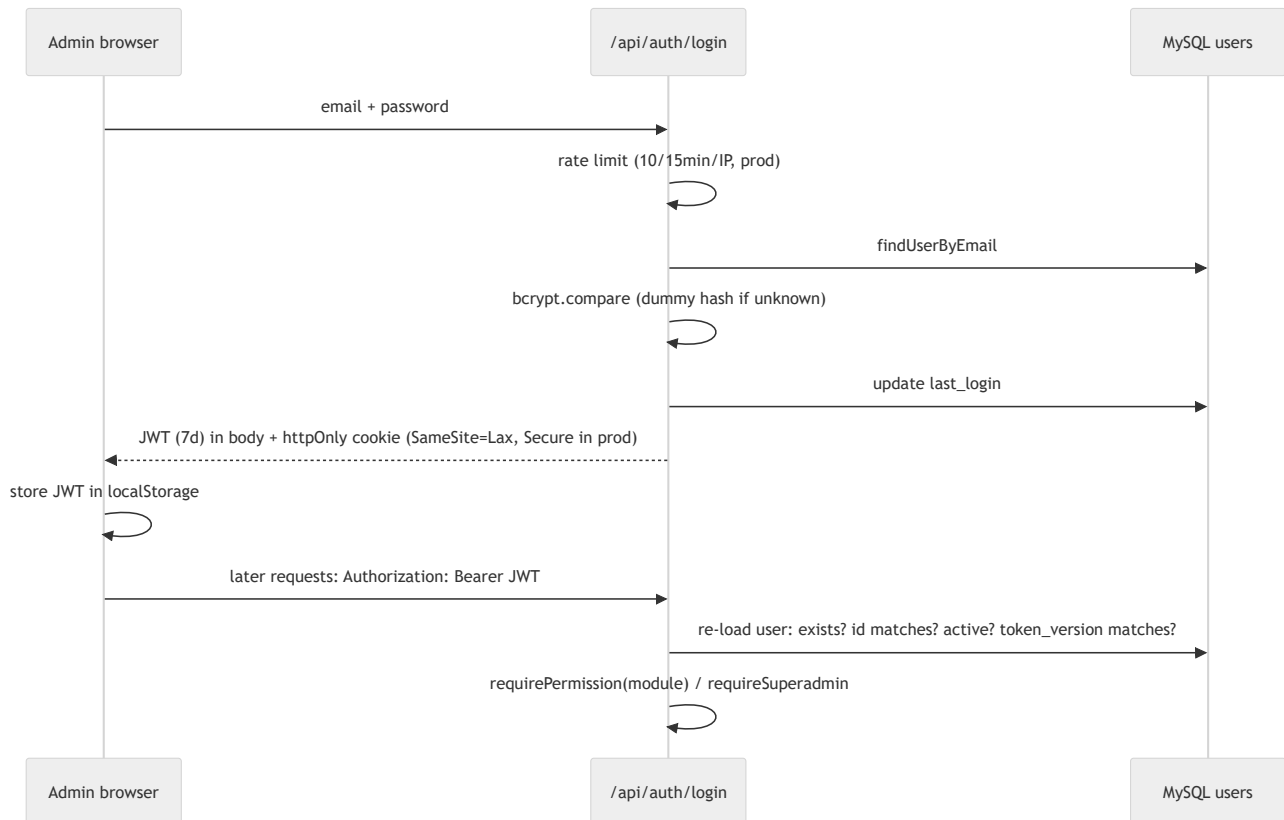
Aspect	Detail	Label
Schema files	001_initial_schema.sql, 002_support_system.sql, 003_full_cms_coverage.sql (untracked)	VERIFIED
Runtime DDL	content_translations table and extras_json columns created by the adapter	VERIFIED mysqlAdapter.ts:1898,1935
Migration strategy	Manual per-file scripts; schema_migrations written, never consulted; server only checks homepage_config exists	VERIFIED
JSON adapter	Dev only; blocked in production by env validation and adapter factory	VERIFIED

Entities and relationships



Singletons: site_settings, contact_settings, homepage_config, video_storyboard, chauffeur_config, why_stay_config, dining_config, global_content. Lists: hero_slides, homepage_sections, villas, gallery_items, facilities, testimonials, video_items, seo_routes, media_assets, experiences, safari_destinations, dining_categories, support_*, content_translations, audit_logs, users.

5. Authentication & Authorisation



- Revocation: `token_version` bumped on logout, disable, enable, password reset and role/status/permission change.
- Roles: `superadmin` (all), `admin` (17 assignable module keys). Max 6 active admins. Last active superadmin protected.

6. Media Storage

Aspect	Detail
Adapter	<code>HostingerMediaStorage</code> (prod) / <code>LocalMediaStorage</code> (dev) — <code>server/storage/*</code>
Location	<code>MEDIA_STORAGE_PATH</code> , default <code><domain>/zanzirangi-media</code> derived from Hostinger <code>hbuilds/versions/<id></code> path; falls back to <code>./uploads</code> if unwritable (that fallback is inside the deploy folder and would be lost on redeploy)
Upload transport	base64 inside JSON (<code>POST /api/admin/media/upload</code>)
Validation	extension allow-list, MIME/extension match, magic-byte signature check, size ≤ <code>MAX_UPLOAD_SIZE</code> MB (default 25), sanitized + randomised filename, resolved-path containment check
Serving	<code>express.static</code> at <code>/uploads</code> with <code>nosniff</code> , <code>sandbox CSP</code> , <code>Cache-Control: public, no-cache</code>
Image processing	none (no resizing/format conversion)

7. Infrastructure

Item	Value	Label
Hosting provider	Hostinger (Node.js web app via hPanel)	INFERRED (code comments <code>server/index.ts:28,36</code> ; <code>env.ts:48-55</code> ; docs)
Domain	<code>zanzirangihouse.com</code>	Configured in code; registration/DNS NOT VERIFIED
SSL / HTTPS	HSTS emitted on API responses in production; <code>Secure</code> cookies in production; certificate NOT VERIFIED	PARTIALLY VERIFIED
CDN	Code comment states Hostinger's CDN rewrites CSP on uploads and caches media	INFERRED (<code>server/index.ts:36-39</code>)
DNS dependencies	NOT VERIFIED	
Port	<code>PORT</code> from platform; default 3000	VERIFIED
Process manager	Hostinger runtime (Passenger referenced in commit <code>adab9b8</code>)	INFERRED
Email infrastructure	NOT FOUND	
Storage	Local disk on host	VERIFIED (code)
Database hosting	Hostinger MySQL on same server (<code>127.0.0.1</code>)	INFERRED (<code>.env.example</code>)
CI/CD	GitHub Actions → GitHub Pages static build only; no pipeline to Hostinger	VERIFIED

8. Build & Release

Step	Command	Output
Client build	<code>vite build</code>	<code>dist/index.html</code> , <code>dist/assets/*</code> (hashed)
Route pre-render	<code>node scripts/generate_routes.js</code>	<code>dist/<route>/index.html</code> with route meta
Server bundle	<code>esbuild server/index.ts --bundle --platform=node --format=esm --packages=external --outfile=server.js</code>	<code>server.js</code> (committed to git)
Package	<code>npm run package:hostinger</code>	<code>hostinger_deploy.zip</code> (<code>dist</code> , <code>server.js</code> , <code>package*.json</code> , <code>public</code> , <code>uploads</code> , <code>.env.example</code>) with secret scan
Deploy	Manual upload + hPanel Node.js setup	—

9. External Integrations

Integration	Mechanism	Server-side?
Google Maps	iframe embed + share link	No
Google Fonts	CSS link	No
WhatsApp	<code>wa.me</code> links	No
Unsplash	image URLs	No

Integration	Mechanism	Server-side?
Gemini (@google/genai)	dependency only — unused	No
Analytics / email / payments / OTA / CRM	NOT FOUND	—

10. Known Architectural Constraints

1. Single process, single host: no horizontal scaling; the in-memory rate limiter is per-process.
2. Chat uses HTTP polling (no WebSockets/SSE).
3. All content is client-rendered; pre-rendered HTML holds metadata only.
4. Media uploads are base64-in-JSON ($\approx 33\%$ overhead, memory-bound).
5. Manual schema migrations; runtime DDL in the adapter.
6. No staging tier; development currently configured against a remote database.